NLP Theory Test [MCQ] (Version : 1)

TEST

Correct Answer

Answered in 2.52 Minutes

Question 1/15

What is the purpose of NLTK in Natural Language Processing (NLP)?

- [ ] To create graphical user interfaces for data visualisation.
- [x] To provide interfaces and libraries for working with human language data.
- [ ] To develop machine learning models for image processing.
- [ ] To simulate human-like conversations using chatbots.

Explanation:

- To develop machine learning models for image processing: This option is incorrect as NLTK primarily focuses on natural language processing tasks, not image processing.
- To create graphical user interfaces for data visualisation: This option is incorrect because NLTK is not specifically designed for creating graphical user interfaces.
- To simulate human-like conversations using chatbots: This option is incorrect because while NLTK can be used to develop chatbots, its primary purpose is not specifically for simulating human-like conversations.

Question 2/15

What is the purpose of removing stop words in text analysis?

- [ ] To identify and remove URLs from the text.
- [ ] To convert text into lowercase format.
- [ ] To highlight the most important words in the text for visualisation purposes.
- [x] To eliminate words that lack significance for search queries.

Explanation:

- To convert text into lowercase format: This option is incorrect because lowercasing is a separate preprocessing step that helps standardise the text data but is not the primary purpose of stop word removal.
- To identify and remove URLs from the text: This option is incorrect because removing URLs is a specific preprocessing step aimed at cleaning text data from web sources, but it is not the primary purpose of stop word removal.
- To highlight the most important words in the text for visualisation purposes: This option is incorrect because stop word removal aims to eliminate common words rather than highlighting important ones.

Question 3/15

In the sentence: "This is an introduction to Natural Language Processing", which words are considered stop words?

- [x] This, is, an, to
- [ ] Is , an
- [ ] This, an, to

- [ ] Is, an, to

Question 4/15

When we want to separate each word in a sentece into a collection of tokens, we need to use:

- [ ] from nltk.tokenize import word_tokens
- [ ] from nltk.stem import PorterStemmer
- [ ] from nltk.stem import WordNetLemmatizer
- [x] from nltk.tokenize import word_tokenize

Question 5/15

What would be the root word if we used the PorterStemmer() as the stemming algorithm on the word "loving"?

- [ ] lovi
- [x] Love
- [ ] Lov
- [ ] Lovely

Question 6/15

N-grams are defined as the combination of N consecutive words. How many bi-grams can be generated from the sentence: "Classification is a great machine learning technique to learn"?

- [ ] 10
- [ ] 9
- [x] 8
- [ ] 7

Question 7/15

Which of the following techniques can be used in the process of converting a keyword into its base form?

- [ ] N-grams
- [x] Lemmatization
- [ ] Cosine Similarity
- [ ] CountVectorizer

Question 8/15

The process of identifying people, locations, or organizations from a given sentence or paragraph is called:

- [ ] Lemmatization
- [x] Named entity recognition
- [ ] Stemming
- [ ] Stop word removal

Question 9/15

Which parameter of CountVectorizer specifies the range of n-values for different n-grams to be extracted?

- [ ] stop_words
- [ ] min_df
- [ ] max_df
- [x] ngram_range

Explanation:

- stop_words: This option is incorrect because the stop_words parameter in CountVectorizer is used to specify a list of words to be ignored during tokenization. It does not specify the range of n-values for n-grams.
- max_df: This option is incorrect because the max_df parameter in CountVectorizer is used to ignore terms that have a document frequency strictly higher than the given threshold. It does not specify the range of n-values for n-grams.
- min_df: This option is incorrect because the min_df parameter in CountVectorizer is used to ignore terms that have a document frequency strictly lower than the given threshold. It does not specify the range of n-values for n-grams.

Question 10/15

Converting text into tokens and then converting them into integers or floats can be done using:

- [ ] TF-IDF
- [ ] Bag of Words

- [ ] Lemmatization
- [x] CountVectorizer

Question 11/15

What is the purpose of tuning the parameters of CountVectorizer?

- [x] To optimise text feature extraction for better model performance.
- [ ] To convert text into lowercase format.
- [ ] To remove stopwords from the text data.
- [ ] To identify and extract meaningful features from the text.

Explanation:

- To convert text into lowercase format: This option is incorrect because converting text into lowercase format is a preprocessing step that ensures consistency in text data but does not directly optimise text feature extraction.
- To remove stopwords from the text data: This option is incorrect because removing stopwords is another preprocessing step aimed at reducing noise in text data but does not involve tuning parameters for optimising feature extraction.
- To identify and extract meaningful features from the text: This option is incorrect because while tuning parameters may indirectly lead to the identification and extraction of meaningful features, the primary purpose is to optimise the process rather than directly identify features.

Question 12/15

What is the main difference between stemming and lemmatization?

- [ ] There is no difference, they are just different python libraries for doing the same thing
- [ ] Stemming comes from the pandas python library while lemmatization comes from NLTK
- [ ] Stem words can be looked up in a dictionary while lemmas cannot
- [x] Stemming can often create non-existent words, whereas lemmas are actual words

Question 13/15

Which one of the following are keyword normalization techniques?
i. Stemming
ii. Part of Speech
iii. Named entity recognition
iv. Lemmatization

- [x] i,iv
- [ ] ii,iii,i
- [ ] iii,ii
- [ ] All of the above

Question 14/15

Which is the correct order for preprocessing text data in Natural Language Processing applications?

- [ ] Lemmatization->remove stop words-

- [ ] >tokenization

- [ ] Tokenization->remove stop words->lemmatization

- [x] Remove stop words->tokenization->lemmatization

- [ ] `Remove stop words->lemmatization->to`

Question 15/15

What does the chi-square statistic measure in the context of text analysis?

- [ ] The proportion of documents in which a term appears.

- [ ] The total number of words in the vocabulary.

- [ ] The frequency of occurrence of words in the text data.

- [x] The difference between observed and expected values of word frequencies across different categories

Explanation:

- The frequency of occurrence of words in the text data: This option is incorrect as it describes the raw frequency of word occurrences, which is not specifically measured by the chi-square statistic.
- The total number of words in the vocabulary: This option is incorrect as the chi-square statistic does not measure the total number of words in the vocabulary but rather the association between word frequencies and categories.

- The proportion of documents in which a term appears: This option is incorrect as it describes the document frequency of terms, which is not directly measured by the chi-square statistic.